

OmTrackVLA on Clearpath Ridgeback

Current two-GPU split, target identity, safety, and measured evidence

RIDGEBACK RTX 5060

YOLO11n + Grounding DINO. Ridgeback CPU tracks identity and owns the 20 Hz motion gate.

NVIDIA THOR

DINOv3 + SigLIP + Qwen3 + OmTrackVLA on Thor CUDA. Its CPU handles the service and cache.

CURRENT STATUS

Local RTX armable trials were observed. The two-GPU hybrid remains dry-run only; its live locked-target motion is unverified.

The hybrid split uses both GPUs

RIDGEBACK COMPUTER | 192.168.131.1

RTX GPU: YOLO11n person boxes; Grounding DINO when searching or recovering.

CPU: BoT-SORT association, optical flow, HSV clothing gallery, ROS sensors, depth position, 75/25 command fusion, 20 Hz safety gate, /cmd_vel.

NVIDIA THOR | 192.168.131.51

Thor GPU: DINOv3, SigLIP, Qwen3, OmTrackVLA waypoint model and visual history.

Thor CPU: TCP/JSON service, JPEG decode, request validation and planner cache.

Interface: project TCP protocol on port 18765; Thor is not a ROS inference node.



Ridgeback to Thor: same JPEG and target only while LOCKED. Otherwise, target metadata without a JPEG.

Thor to Ridgeback: waypoints and a matching target result, checked before command fusion.

Perception finishes before the Thor planner request

1 SELECT

Ridgeback selects a compressed RGB frame for the target loop (up to 3 Hz).

2 FIND

RTX YOLO detects all people. Grounding DINO checks the prompt attribute when due.

3 VERIFY

CPU BoT-SORT and clothing features keep the same person; only LOCKED is valid.

4 PLAN

For LOCKED, Thor receives the same JPEG and target. Its planner updates at up to 2 Hz; cached paths have a 0.75 s age limit.

5 CONTROL

Ridgeback checks reply identity and freshness, localizes depth, fuses direction, and runs the 20 Hz gate.

Current prompt: "Follow the person who is wearing black T-shirt." Grounding checks the garment; the HSV signature describes visible clothing. Recovery checks both prompt and appearance.

Ridgeback makes the final motion decision

The locked leader and the model path meet in the local controller.

LEADER DIRECTION

Ridgeback projects the locked person's box into a depth-measured position. The controller aims toward that position while respecting a 0.9 m follow distance.

MODEL DIRECTION

Thor's OmTrackVLA predicts eight local waypoints. Waypoint index 1 supplies the raw direction; the model does not report who it is following.

DIRECTION FUSION ON RIDGEBACK CPU

When the model direction agrees with the leader, the controller blends about **75% leader direction and 25% aligned model direction**. An opposing or sideways model proposal is replaced by the leader direction. This ratio describes command fusion, not the share of GPU computation.

Every 50 ms: dry-run, deadman, E-stop, camera/model/scan freshness, LOCKED target, depth, LiDAR clearance, speed and acceleration checks. Failure commands zero or stops publication.

What the September 28-29 checks establish

Full-Thor live dry-run	187 frames; 93 LOCKED. The planner updated 31 times over 30.241 s of lock (~1.03 Hz under a 2 Hz cap). E-stop stayed active.
Hybrid offline replay	90 current-camera blue-basket frames; 49 LOCKED. All five compared target/trajectory decision fields matched the earlier full RTX replay.
Hybrid transfer	41 invalid-target replay frames sent target metadata to Thor without a JPEG. Ridgeback perception used about 2,390 MiB RTX memory.
Hybrid live searching	46 no-person frames at 3.054 Hz. Bridge pipeline median/p95: 25.564/27.673 ms. 20 Hz control interval maximum: 51.455 ms.
Service failures	Stopping either service cleared the prediction in separate dry-runs. Deadman was unheld, so physical stopping was not measured.
Tests	66 project tests passed after the hybrid implementation. Hybrid locked-target live planner cadence and motion have not passed validation.

These runs used different scenes and timings. They do not show a hybrid speedup over full RTX or full Thor.

Measurements and limits: docs/COMPUTE_PLACEMENT_PLAN.md and docs/PROJECT_CONTEXT.md

Four supervised runs, four different conditions

Each window lasted 39.5 s; target and gate counts are from 80 status samples at 2 Hz.

A 25% model / 75% target	Blue-basket prompt. LOCKED 78/80; gate ready 54/80. Wheel-odometry endpoint change 1.26 m. A stale LiDAR scan blocked commands in 21/80 samples.
B 75% model / 25% target	Black T-shirt prompt. LOCKED 4/80; target-not-locked gate 76/80. Endpoint change 0.109 m. Appearance recovery repeatedly lost identity, so the blend rarely reached motion.
C 100% model / 0% target	Black T-shirt prompt. LOCKED 68/80; gate ready 64/80. Endpoint change 2.17 m. The aligned model direction applied in 26 outside-follow-distance samples.
D 0% model / 100% target	Black T-shirt prompt. LOCKED 80/80; obstacle gate 26/80. Endpoint change 4.26 m, but 0.63 m occurred during a 9.45 s controller-command gap; joystick use is possible. A fresh planner result and model speed cap still applied.

These percentages blend direction only for aligned proposals. Target lock, LiDAR and the 0.9 m follow distance still gate motion.

Prompts, lock duration, starting pose and scene differed. Odometry is not surveyed travel. Run D's movement cannot be attributed to this controller without the command-mux inputs and output. The runs do not rank the weights or measure target-selection accuracy.

Hybrid remains a dry-run launch

Thor terminal

```
ssh robot@192.168.131.51; cd /home/robot/dev/omtrackvla; ./real_robot/start_planner_thor.sh
```

Ridgeback terminal

```
cd /home/robot/Desktop/omtrackvla; set ROBOT_NAMESPACE=r100_0160, CAMERA_TOPIC=camera/color/image_raw/compressed, CAMERA_COMPRESSED=true; run ./real_robot/start_ridgeback_hybrid.sh
```

RViz

On Ridgeback: ./real_robot/start_ridgeback_rviz.sh in a separate terminal.

Validation before hybrid motor output

- Capture a live black T-shirt LOCKED hybrid dry-run and measure Thor planner updates, target IDs, and stale frames.
- Compare full RTX, full Thor, and hybrid with the same scene; record p50/p95/max stage times, GPU load/memory, CPU/network load, and 20 Hz deadlines.
- Record all twist_mux inputs and output, motor feedback, odometry and gate decisions in the next local RTX armable trial; test hybrid disconnect/reconnect before any arming change.

The local full RTX supervised launcher remains the only validated armable path. The hybrid launcher rejects OMTRACKVLA_ARM_OUTPUT=1.